

Module 2

Session 3

1. Declare variables for a Flipkart product: productName (as a string), price (float), and rating (double). Assign sample values and print each variable with its data type.

```
#include <stdio.h>

main()
{
    char prdt[] = "TV";
    float price = 199998.9f;
    float rate = 4.9;
    printf("Product Name = %s\n", prdt);
    printf("Data Type = char[] (string)\n");
    printf("Sizeof = %zu bytes\n\n", sizeof(prdt));
    printf("Price = Rs.%.2f\n", price);
    printf("Data Type = float\n");
    printf("Sizeof = %zu bytes\n\n", sizeof(price));
    printf("Rate = %.1f\n", rate);
    printf("Data Type = float\n");
    printf("Sizeof = %zu bytes\n", sizeof(rate));
}
```

**2. Create a constant variable to store the GST rate (for example, 18%) and use it to calculate the final price of a Zomato order with a given base price.

Constraint: The GST rate must not be changeable after its initial assignment.**

```
#include<stdio.h>

main()
{
    const float GST_RATE = 18.0;
    float basePrice, gstAmount, finalPrice;
    printf("Enter the base price of the Zomato order: ");
    scanf("%f", &basePrice);
    gstAmount = (basePrice * GST_RATE) / 100;
    finalPrice = basePrice + gstAmount;
    printf("\nGST Rate = %.2f%%", GST_RATE);
}
```

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```
printf("\n\nBase Price = Rs. %.2f", basePrice);  
printf("\nGST Amount = Rs. +%.2f", gstAmount);  
printf("\n\n-----");  
printf("\nFinal Price = Rs. %.2f", finalPrice);  
}
```

3. Write a program that stores your favorite Spotify playlist's name (string), total number of songs (int), and average song duration in minutes (float). Print all values in a single formatted sentence.

```
#include<stdio.h>  
  
main()  
{  
    char playlist[50];  
    int songs;  
    float duration;  
    printf("Enter Playlist Name: ");  
    scanf(" %[^\\n]", playlist);  
    printf("Enter Total Number of Songs: ");  
    scanf("%d", &songs);  
    printf("Enter Average Song Duration (minutes): ");  
    scanf("%f", &duration);  
  
    printf("\nMy favorite Spotify playlist \"%s\" contains %d songs with an average duration  
of %.2f minutes per song.", playlist, songs, duration);  
}
```

**4. Identify and correct the invalid variable names from this list: 1stPlayer, player_age, \$score, total-marks, userName.

Hint: Apply standard variable naming rules for your programming language.**

Invalid Variable Names and Their Corrections: -

1. 1stPlayer → Invalid (variable names cannot start with a number)
Corrected: - firstPlayer or player1
2. player_age → Valid
3. \$score → Invalid (contains the special character \$)
Corrected: score

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4. total-marks → Invalid (contains the special character -)

Corrected: total_marks or totalMarks

5. userName → Valid

Final Answer: - Invalid variable names: -

- 1stPlayer → firstPlayer
- \$score → score
- total-marks → total_marks

Valid variable names: -

- player_age
- userName